O → mai mic sau egal
o → mai mic
W → mai mare sau egal
w → mai mare
Theta → egal

f(n) = O(g(n))

exista un c, un n0, astfel incat oricare ar fi n >=n0:

f(n) <= c * g(n)

lim f(n)/g(n) < inf
n→ inf

f(n) = W(g(n))

exista un c, un n0, astfel incat oricare ar fi n >= n0:

f(n) >= c * g(n)

lim f(n)/g(n) > 0
n→ inf

n^2 = O(n^3)
n^2log n = O (n^3)
3n^3 + 5 n^2 + 15 n^2 log n = O(n^3)

n^3 = W (n^2)
n^3 = W(n^2 log n)

3n^3log^2n = W(n ^3)

f(n) = o(g(n))

oricare ar fi c, exista un n0(c) a.i. oricare ar if n>=n0(c):

f(n) < c * g(n)

lim f(n)/ g(n) = 0
n→ inf

n^2 = o(n^3)
n^log^2n = o(n^3)

f(n) =w (g(n))

oricare ar fi c, exista un n0(c) a.i. oricare ar fi n>= n0(c)

f(n) > c* g(n)

lim f(n)/ g(n) = inf
n→ inf

f(n) = theta (g(n))

exista un c1,c2, n0 a.i. oricare ar fi n >= n0

c1 * g(n) <= f(n) <= c2 * g(n)

lim f(n)/ g(n) apartine R+ (apartine (0,inf))
n→ inf

n^2 + n = O(n^2)

exista un c, n0 a.i. oricare n>= n0

n^2 + n <= c * n^2

alegem c =2 :

n^2 + n <= 2 * n^2 => adevarat oricare n >=1 => n^2 + n = O(n^2)

lim (n^2+n)/n^2 = 1 < inf => n^2+n = O(n^2)
n→ inf

n^2+n != o(n^2)

oricare c, exista un n0(c) a.i. oricare n>= n0(c)

n^2 + n < c * n^2

contrazic definitia astfel:

exista un c, oricare ar fi n0 , exista n >= n0

n^2 + n >= c * n^2

c = 1 => n^2 + n >= n^2, adevarat pentru orice n >= 1

=> n^2 + n != o (n^2)

lim $n \to \inf$ $(n^2+n)/n^2 = 1 != 0 \Rightarrow n^2+ n != o(n^2)$

$n^2 + n^3 = \text{Theta}(n^3)$

exista c1,c2,n0 a.i. oricare n >= n0

$c1 * n^3 <= n^2 + n^3 <= c2 * n^3$

c1 =1, c2 =2 =>

$n^3 <= n^2 + n^3 <= 2 n^3$ => adevarat pentru orice n>= 1 =>

$n^2 + n^3 = \text{Theta}(n^3)$

lim $n \to \inf$ $(n^2 +n^3)/ n^3 = 1$ apartine R+ => $n^2 + n^3 = \text{Theta}(n^3)$

$f(n) = O(g(n)) <=> g(n) = W(f(n))$

=>

f(n) = O(g(n)) =>
exista c, n0 a.i. oricare n >= n0

$f(n) <= c* g(n) \Rightarrow g(n) >= (1/c) * f(n)$

notez 1/c cu c1 =>

g(n) >= c1 * f(n), oricare n >= n0 => g(n) = W(f(n))

lim $n \to \inf$ $f(n)/ g(n) < \inf$

lim $n \to \inf$ $f(n)/g(n) = x \Rightarrow$ lim $n \to \inf$ $g(n) / f(n) = 1/x > 0$

<=

g(n) = W(f(n)) =>
exista c, n0 a.i. oricare n> n0

$g(n) >= c * f(n) \Rightarrow f(n) <= (1/c) * g(n)$

notez 1/c = c1 => $f(n) <= c1 * g(n)$ oricare n >= n0 => f(n) = O(g(n))

f(n) = Theta(g(n)) <=> f(n) = O(g(n)) si f(n) = W(g(n))

=>

f(n) = Theta(g(n)) =>
exista c1,c2, n0 a.i. oricare n>=n0

c1*g(n) <= f(n) <= c2*g(n)

c1*g(n) <= f(n) => f(n) >= c1*g(n) => f(n) = W(g(n))

f(n) <= c2*g(n) => f(n) = O(g(n))

<=

f(n) = O(g(n)) =>
exista c1, n01 a.i. oricare n>= n01

f(n) <= c1 *g(n)

f(n) = W(g(n)) =>
exista c2, n02 a.i. oricare n>= n02

f(n) >= c2*g(n)

c2 * g(n) <= f(n) <= c1 * g(n), oricare ar fi n >= n0 = max(n01,n02)